Adding a node is as simple as booting up Elasticsearch on a different machine but in the same network with the same `cluster.name` (a property in the elasticsearch.yml file), provided security is disabled.

> **WARNING** Since version 8.0, by default, Elasticsearch installations enable security, where `xpack.security.enabled` is `true`. When you bring up the Elasticsearch server for the first time, it generates the required keys and tokens and instructs you about the steps to be followed to obtain successful Kibana connectivity. If you are experimenting with Elasticsearch on your local machine, you can disable security, but I strongly recommend not jumping into production with an unsecured setup by setting the property `xpack.security.enabled` to `false` in the elasticsearch.yml file. An unsecured setup is very dangerous, and you are asking for trouble.

Bringing up a second node helps Elasticsearch create the replica shard on it. As figure 14.4 shows, Elasticsearch instantly creates the replica 1 shard, which is an exact copy of shard 1, when the second node starts and joins the cluster. The contents of shard 1 are synced up to replica 1 immediately, and once they are in sync, any write operations on shard 1 are replicated to replica 1 going forward. The same applies for multiple shards and multiple replicas.

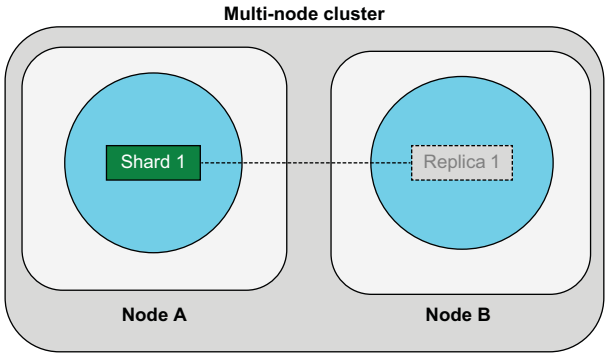


Figure 14.4 The replica shard is created on a second node.

If we add more nodes to the cluster, Elasticsearch scales the cluster elegantly with additional nodes. It automatically distributes the shards and replicas with the addition (or removal) of nodes. Elasticsearch manages everything transparently behind the scenes such that we—ordinary users or administrators—don't have to worry about the mechanics of communication between nodes, how the shards and their data are distributed, and so forth. Figure 14.5 illustrates how a shard (shard 2) is moved to a newly joined second node (thus making a multi-node cluster) and replicas are created.